LEVEL UP ACADEMY

Class 9 | GSEB | Science

Annual Exam Practice Paper 1 (GSEB pattern)

Chapter / Topic: Motion · Force and Laws of Motion · Work and Energy · Sound · Matter in Our Surroundings · Is Matter Around Us Pure? · Structure of the Atom · The Cell · Tissues

Name: _____ Roll No.: _____ Date: _____

Time: 3 hours **Max. Marks: 80**

General instructions:

1. Section A (Q1–24) is compulsory: 1 mark each.
2. Section B: answer any 9 of 12 questions (2 marks each). Section C: answer any 6 of 8 (3 marks each). Section D: answer any 5 of 7 (4 marks each).
3. Show formulae and units in numericals, and draw neat, labelled diagrams wherever needed.
4. This paper is topic-based and follows the reported GSEB blueprint (A 24 · B 18 · C 18 · D 20). Check the topic list and blueprint against your school's annual exam before using it.

SECTION A (Q1–8): CHOOSE THE CORRECT OPTION — 8 marks

Q1. A body moves with uniform velocity. Its acceleration is: **[1]**

(a) zero (b) positive
(c) negative (d) always increasing

Q2. One newton is equal to: **[1]**

(a) 1 kg m/s (b) 1 kg m/s^2
(c) 1 g cm/s^2 (d) 1 kg m^2/s^2

Q3. If the speed of a moving body is doubled, its kinetic energy becomes: **[1]**

(a) half (b) double
(c) three times (d) four times

Q4. Which of these is a heterogeneous mixture? **[1]**

(a) sand in water (b) air
(c) salt solution (d) brass

Q5. The direct change of a gas into a solid, without becoming a liquid, is called: **[1]**

(a) sublimation (b) condensation
(c) evaporation (d) deposition

Q6. The maximum number of electrons that the M shell can hold is: **[1]**

(a) 2 (b) 8
(c) 18 (d) 32

Q7. Which of these is found in plant cells but not in animal cells? **[1]**

(a) mitochondria (b) plastids
(c) Golgi apparatus (d) ribosomes

Q8. Bats can fly and find their food in the dark because they use: **[1]**

(a) infrasound (b) light
(c) ultrasound (d) magnetism

SECTION A (Q9–14): FILL IN THE BLANKS — 6 marks

Q9. The area under a velocity–time graph gives the _____ of the object. **[1]**

Q10. The SI unit of power is the _____. **[1]**

Q11. Sound waves in air are _____ waves. **[1]**

Q12. The temperature at which a solid melts at atmospheric pressure is called its _____. **[1]**

Q13. Rutherford's experiment showed that most of the space inside an atom is _____. **[1]**

Q14. Xylem and phloem are examples of _____ permanent tissues. **[1]**

SECTION A (Q15–19): STATE TRUE OR FALSE

5 marks

Q15. An object can have zero displacement even after covering a large distance. **[1]**

Q16. Action and reaction forces act on the same body. **[1]**

Q17. Evaporation takes place only at the boiling point of a liquid. **[1]**

Q18. Isotopes of an element have different atomic numbers. **[1]**

Q19. Lysosomes contain digestive enzymes. **[1]**

SECTION A (Q20–24): ANSWER IN ONE WORD

5 marks

Q20. Name the quantity equal to the product of mass and velocity. **[1]**

Q21. What is the SI unit of frequency? **[1]**

Q22. Name the method used to separate cream from milk. **[1]**

Q23. Who discovered the neutron? **[1]**

Q24. Name the tissue at the tips of roots and stems that makes them grow longer. **[1]**

SECTION B

answer any 9 of 12 · 18 marks

Answer any 9 questions. 2 marks each.

Q25. Give two differences between uniform and non-uniform motion, with one example of each. **[2]**

Q26. A car travels 120 km in 2 hours and then 60 km in the next 1 hour. Find its average speed for the whole journey. **[2]**

Q27. State Newton's first law of motion. Why is it also called the law of inertia? **[2]**

Q28. What force is needed to give a car of mass 1500 kg an acceleration of 2 m/s^2? What acceleration would the same force give a 3000 kg truck? **[2]**

Q29. When is the work done by a force zero? Give one example. **[2]**

Q30. What is an echo? Why do we not hear an echo in a small room? **[2]**

Q31. What is sublimation? Name two substances that sublime. **[2]**

Q32. Why are cotton clothes more comfortable than synthetic clothes in summer? **[2]**

Q33. Give two differences between a true solution and a colloid. **[2]**

Q34. Write the electronic configurations of magnesium (Z = 12) and chlorine (Z = 17), and give the valency of each. **[2]**

Q35. Name the cell organelle that: (a) makes proteins, (b) carries out photosynthesis, (c) stores starch in plant cells, (d) helps form lysosomes. **[2]**

Q36. What are stomata? Give two of their functions. **[2]**

SECTION C

answer any 6 of 8 · 18 marks

Answer any 6 questions. 3 marks each.

Q37. Derive the equation $v^2 - u^2 = 2as$ using a velocity–time graph. **[3]**

Q38. A ball of mass 0.5 kg moving at 10 m/s is hit straight back by a bat at 15 m/s. The bat is in contact with the ball for 0.01 s. Find the change in momentum of the ball and the average force exerted by the bat. **[3]**

Q39. A girl of mass 40 kg climbs a staircase 5 m high in 10 s. Find the work she does against gravity and her power. ($g = 10\ m/s^2$) **[3]**

Q40. Explain how a vibrating tuning fork produces sound, and how the sound travels through air to our ears. **[3]**

Q41. How will you separate: (a) common salt from a mixture of salt and sand, (b) acetone from a mixture of acetone and water, (c) the different gases of the air? **[3]**

Q42. What are canal rays? Who discovered them, and which particle did they lead to? **[3]**

Q43. Draw a neat, labelled diagram of an animal cell. **[3]**

Q44. Give one difference each in the structure of parenchyma, collenchyma and sclerenchyma, and one function of each. **[3]**

SECTION D

answer any 5 of 7 · 20 marks

Answer any 5 questions. 4 marks each.

Q45. (a) What is uniform circular motion? Why is it called accelerated motion?
(b) An athlete runs once around a circular track of radius 70 m in 44 s. Find his speed. (Take $\pi = 22/7$.) **[4]**

Q46. (a) State the law of conservation of energy.
(b) A 2 kg ball is dropped from a height of 10 m. Find its potential energy and kinetic energy at the top, at a height of 5 m, and just before it hits the ground. ($g = 10\ m/s^2$) What do you notice? **[4]**

Q47. (a) What is SONAR?
(b) A ship sends an ultrasound signal to the sea bed and receives its echo 2 s later. If sound travels at 1500 m/s in sea water, how deep is the sea there?
(c) Give one more use of ultrasound. **[4]**

Q48. (a) What is the Tyndall effect? Give one everyday example.
(b) Compare a solution, a suspension and a colloid on: (i) whether the particles can be seen, (ii) whether they settle down. **[4]**

Q49. (a) How can a gas be changed into a liquid?
(b) Why is solid carbon dioxide called dry ice?
(c) Why do water droplets form on the outside of a glass of iced water? **[4]**

Q50. (a) Write two postulates of Bohr's model of the atom.
(b) An oxygen atom has atomic number 8 and mass number 16. Draw its atomic structure, and give its electronic configuration and valency. **[4]**

Q51. (a) Draw a neat, labelled diagram of a neuron.
(b) Give the function of the dendrites and of the axon. **[4]**

~ All the best for your annual exam! ~